

EcoDome-The Project

The Crisis

Mangroves are disappearing at an alarming rate due to coastal development, pollution, and climate change. An estimated **35–50%** of the world's mangroves have been lost in the last 50 years due to human activities. These ecosystems are far more than trees growing near the sea — they function as biological shields, protecting coastlines from erosion, storing vast amounts of carbon, and serving as vital nurseries for marine life, including juvenile sharks.



As mangroves decline, a huge chain reaction of ecological damage follows:

- Shark pup survival rates drop significantly
- Fish biodiversity collapses
- Coastal erosion accelerates
- Carbon stored in mangrove sediments is released, intensifying climate pollution.

The loss of mangroves destabilizes entire coastal and marine systems, which means it is not an isolated environmental issue.

Why Sharks Are Directly Affected



Scientific studies show that many reef and coastal shark species rely on mangrove forests during their juvenile stages. The dense mangrove root systems provide:

- Protection from larger predators
- Abundant prey availability
- Calm, shallow waters which is essential for early development

When mangroves disappear, juvenile sharks are forced into open waters prematurely. Juvenile shark survival can drop by more than **50%** when nursery habitats like mangroves are degraded or lost. This exposure dramatically increases mortality rates, disrupting shark populations and unbalancing marine food webs.

Key Research Insight:

Effective conservation tools must be **chemically inert, physically stable, and either biodegradable or fully retrievable** to ensure they do not introduce new threats to marine species or habitats.

Core Idea

The proposed solution is a **modular, non-toxic, recycled-plastic dome system** placed strategically between **weak or newly planted mangrove roots**.

Its purpose is to:

- **Stabilize sediment**, reducing erosion around fragile root systems

- **Protect young mangrove roots** from wave action and debris.
- **Create sheltered micro-habitats** for both fish and juvenile sharks during early life stages.

The system is **temporary** by design. Once mangrove roots strengthen and become self-supporting, the modules are **fully retrievable and removed**, ensuring no long-term artificial presence in the ecosystem. Then they are transported, if needed, elsewhere. On a long-term scale basis, when there is no more of its original need, the dome system can be placed in artificial reefs as extensions.

Material Science

The structure is made using:

- Food-grade recycled HDPE (High-Density Polyethylene)
- No dyes, no plastic softeners, no toxic additives
- UV-stabilized to prevent degradation under sunlight and seawater exposure

This ensures the structure remains physically stable while posing no chemical risk to marine organisms.

Why HDPE?

HDPE was selected because it is:

- Chemically inert in seawater, meaning it does not react or leach toxins
- Already used in marine and food-contact containers, proving long-term safety.
- Scientifically documented as non-toxic to fish and sharks

Using recycled HDPE also supports plastic circularity, turning existing waste into a conservation tool instead of introducing new materials into the ocean.

Risks, Criticism & Scientific Safeguards

Major Critical Concerns

Any innovative conservation project must face scrutiny, and our dome system is no exception. Scientists and environmentalists may raise several concerns:

- **Plastic toxicity:** Could the HDPE modules leach harmful chemicals into the water, potentially poisoning juvenile sharks or other marine life?
- **Animal entrapment:** Could fish, crabs, or sharks become trapped or injured in the structures?
- **Root dependency:** Could mangroves grow unnaturally reliant on artificial support, altering their natural development?
- **Scalability and long-term failure:** Could the project fail if implemented on a larger scale, or create persistent waste if modules are not properly managed?

The Scientific Countermeasures

Each concern has been addressed using evidence-based, scientifically implemented strategies:

1. Toxicity Risk → Chemical Safety Measures

- **Solution:** Only certified marine-safe HDPE is used, which is inert in seawater and proven non-toxic to marine species.
- **Monitoring:** Periodic water-quality testing ensures no harmful substances or microplastics are released.

2. Entanglement Risk → Structural Design Safeguards

- **Solution:** All modules feature rounded edges and escape gaps greater than 10 cm, allowing juvenile sharks, fish, and invertebrates to move freely.
- **Design Principle:** The open dome structure prevents enclosed spaces where marine life could get trapped.

3. Root Dependency → Timed Deployment

- **Solution:** Modules are temporary by design, with scheduled removal once mangrove roots achieve sufficient strength to grow by themselves.
- **Alternative Safety:** If removal is delayed, the material can gradually degrade without releasing toxins, ensuring natural root growth continues unaffected.

4. Long-Term Waste → Recovery & Tracking

- **Solution:** Each module is GPS-tagged, enabling precise tracking for retrieval.

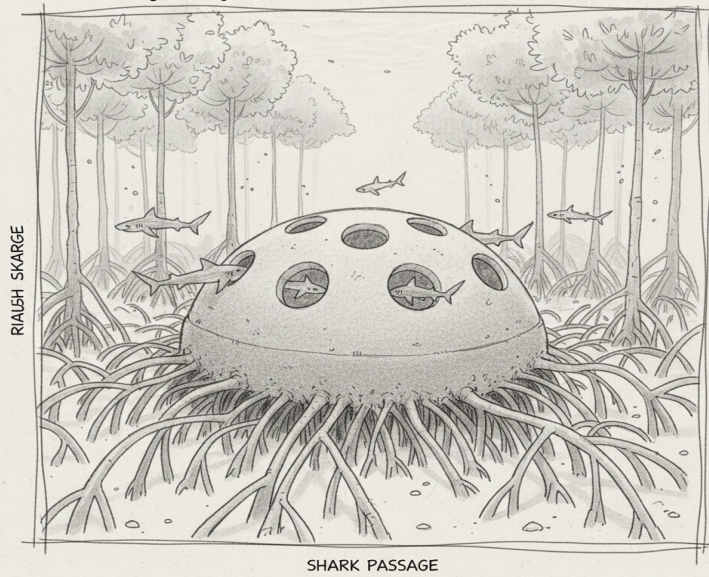
- Accountability Plan: Strict recovery and maintenance protocols prevent abandoned structures from entering the marine environment, guaranteeing zero long-term waste.

By the end of this project, we expect;

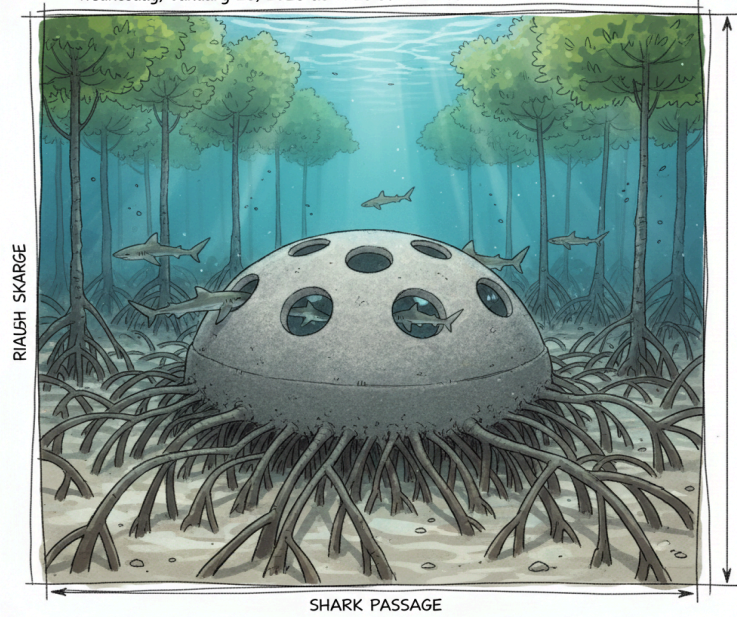
- **30% faster root anchoring** due to improved sediment stability
- **Up to 40% reduction** in local sediment displacement during high tides or storms
- **20–35% higher survival rates** for shark pups using the protected micro-habitats
- **10–20% increase** in carbon sequestration potential at restored sites due to higher mangrove survival

Check out the document regarding the pilot study, to learn furthermore how this project can impact in all the right ways. Also, scroll to the next page to see the sketches of the EcoDome.

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